

DiagnoVision

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Chapter 1

Introduction

DiagnoVision is designed to provide accessible healthcare solutions for common, non severe skin diseases. Today with so much advancement in technology, individuals are still struggling with limited access to medical professionals, leading to delays in diagnosis and unnecessary doctor visits for minor ailments. Now with the rapid advancements in AI and computer vision, our system leverages these technologies to offer accurate health insights through a user-friendly application. DiagnoVision will empower users by allowing them to input symptoms, images, and receive actionable recommendations, reducing the need for physical consultations and improving overall healthcare efficiency. [1].

1.1 Problem Statement

The reason for the development of this software is to address the challenges of timely and accessible healthcare, particularly for common, non-severe skin illnesses that can often be managed without a doctor. Even today, many individuals still lack access to reliable healthcare leading to delayed treatments and unnecessary visits to doctors for minor conditions. So here comes DiagnoVision that aims to bridge this gap by providing a platform where users can input symptoms and images to receive accurate, actionable health insights. By using AI and computer vision, DiagnoVision will analyze the provided data, predicts potential diseases, and offers recommendations for treatment, which will minimize the need for in-person consultations. Moreover, real-time data on the spread of diseases in specific regions and cities will empower users to take hygienic measures, which will reduce the risk of infection. DiagnoVision will also recommend nearby doctors for further consultation if necessary. Designed with a user-friendly interface, our solution ensures accessibility for all, enabling even non-experts to use the system effectively and confidently manage their health.

1.2 Scope

The DiagnoVision project aims to develop a medical chatbot model for skin disease detection for three diseases which are Acne, Eczema, and Psoriasis and takes text description and skin images as inputs. This model's primary functionality is diagnosis and medical advice. The users are asked about their symptoms and even upload photos of skin conditions they might be suffering from so that the chatbot can help them determine the skin condition that they might have. Besides diagnosis, DiagnoVision has two secondary features: monitor the rapid spread of diseases within the user's area and recommendation of skin care depending on the current weather. To a greater extent, the chatbot offers medical help to the users regarding specific skin diseases. The elements of this project are as follows: Data input system, skin diseases detection models, real-time disease spread information, skincare advice based on the weather. The scope of this project is implementation of Natural Language Processing (NLP) and Computer Vision (CV) models, to offer diagnosis and medical advice for three diseases. The project will concentrate on improving user experience because of the improved and easy layouts that can be accessed in the mobile phones. The system will be checked and validated against standard diagnostic protocols with provisions for modifications depending on the feedback and the improvements in technology.

1.3 Modules

DiagnoVision is structured into several key modules that perform as a medical chatbot. These modules are listed below:

1.3.1 Module 1

In this module, the user's textual description of the symptoms is inputted into Natural Language models to determine if the user has a certain skin condition. The text module will look for the symptoms in the text and will match the symptoms for a specific disease. It serves as the diagnostic tool and works well with the image analysis module to come up with a better diagnosis.

1. Extract and analyze key symptoms from user input
2. Match text-based symptoms with known conditions (Acne, Eczema, Psoriasis).

1.3.2 Module 2

This module analyzes images of skin areas uploaded by users and classifies whether it is affected or not. If affected, it identifies visual indicators of the three targeted skin conditions using CV models.

1. Process and enhance images for accurate analysis
2. Detects visual patterns indicative of Acne, Eczema, or Psoriasis.

1.3.3 Module 3

This module combines the results from both the Textual model and Image Analysis modules to provide an accurate diagnosis. It uses a decision-making algorithm to combine the results from both sources and offer the most accurate diagnosis possible.

1. Use the findings elicited from text scan and image scan for diagnosing.
2. Implement decision making algorithms which helps in assessment of the inputs from the two modules.
3. Provide the final diagnosis and confidence score for the diagnosed condition

1.3.4 Module 4

This module keeps track of the rapid spread of diseases in the user's area, helping them be aware of disease spread in their region and take precautionary measures. It uses data from public sources to identify and report disease spread

1. Get Information from local data sources for rapid spread of diseases.
2. Provide alerts and updates on rapidly spreading conditions in the user's vicinity
3. Adjust diagnostic recommendations based on localized data.

1.3.5 Module 5

This module delivers personalized skincare advice based on current weather conditions. It integrates weather data APIs to offer tips that help users take measures for their skin health.

1. Retrieve and analyze real-time weather data
2. Provide tailored skincare recommendations for current weather conditions.
3. Update advice dynamically as weather conditions change.

1.3.6 Module 6

This module provides the user interface and manages the overall user experience within the app

1. User-friendly Interface: Easy to use, simple navigation and layout for easy access to all features.
2. Data Input: Allows users to input symptoms as text and upload images of their skin.
3. Feedback and Notifications: Delivers diagnostic results, recommendations, and alerts to users.

1.4 User Classes and Characteristics

The user classes and their description is given below:

User class	Description
General User	A General User seeks to diagnose skin conditions like Acne, Eczema, or Psoriasis using the DiagnoVision app. Around 500 users are expected to input symptoms or images for AI analysis and receive medical advice, mostly through smartphones.
Healthcare Institution	Hospitals or clinics integrate DiagnoVision into their workflow to streamline patient triage for minor skin conditions, optimizing healthcare delivery with quick AI-based diagnoses.
Developers	Developers are responsible for building, maintaining, and enhancing the DiagnoVision system. Their tasks include optimizing AI and machine learning models for better diagnostic accuracy, fixing bugs, and ensuring smooth integration between the front-end mobile app and back-end services. They also monitor system performance and implement feature updates based on user feedback and advancements in AI technology.
Medical Data Provider	Medical Data Providers offer essential datasets for training and refining the AI models used in DiagnoVision. These datasets, often sourced from reliable medical institutions, are critical for ensuring accurate diagnoses of skin conditions. Providers also collaborate with developers to ensure that the data remains up-to-date, secure, and compliant with healthcare standards, supporting the continuous improvement of diagnosis accuracy.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The selection of the requirement gathering technique(s) will depend on the type of project. For instance, • Use case (use case diagram + detail use case) is an effective technique for interactive end-user applications. • Event- response table is for real-time system in which most of the functionalities are performed at backend. • Storyboarding for graphically intensive applications.

Create the suitable representation here.

2.2 Functional Requirements

This section outlines the functional requirements of DiagnoVision. Each module's functional requirements are expressed in natural language and organized by system feature.

2.2.1 Module 1: Text-Based Symptom Analysis

Following are the requirements for module 1:

1. The system shall allow the user to input a description of their skin-related symptoms via a text field.
2. The system shall analyze the text input using Natural Language Processing (NLP) to identify key symptoms.

3. The system shall match identified symptoms to known conditions (Acne, Eczema, Psoriasis) using a predefined symptom-disease mapping.
4. The system shall provide the user with a preliminary diagnosis based on the text input.
5. The system shall generate a confidence score for the diagnosis based on the accuracy of symptom matching.
6. The system shall suggest additional symptoms to the user for further clarification, if needed.

2.2.2 Module 2: Image-Based Skin Analysis

Following are the requirements for module 2:

1. The system shall allow the user to upload images of affected skin areas.
2. The system shall process and enhance the uploaded images for better feature extraction.
3. The system shall analyze the images using a Computer Vision (CV) model trained to detect Acne, Eczema, and Psoriasis.
4. The system shall classify the image as either affected or unaffected by any of the three skin conditions.
5. The system shall provide a visual diagnosis along with a confidence score for each image.
6. The system shall allow the user to upload multiple images for a more comprehensive diagnosis.

2.2.3 Module 3: Decision-Making and Diagnosis Combination

Following are the requirements for module 3:

1. The system shall integrate the results from both the text-based analysis and the image-based analysis.
2. The system shall use a decision-making algorithm to reconcile conflicting results from the two sources.

3. The system shall calculate a final diagnosis based on the combined results from both analyses.
4. The system shall provide a confidence score for the combined diagnosis.
5. The system shall update the diagnosis dynamically if the user modifies their inputs or uploads additional images.

2.2.4 Module 4: Disease Spread Monitoring

Following are the requirements for module 4:

1. The system shall retrieve local disease spread data from public sources, such as health agencies or open data APIs.
2. The system shall alert users if a disease is spreading rapidly in their area.
3. The system shall provide precautionary measures based on the diseases spreading in the user's area.
4. The system shall update diagnostic recommendations based on regional data about disease spread.

2.2.5 Module 5: Personalized Skincare Advice Based on Weather

Following are the requirements for module 5:

1. The system shall retrieve and analyze real-time weather data using a weather API.
2. The system shall provide personalized skincare advice based on the current weather conditions (e.g., UV index, humidity, temperature).
3. The system shall update skincare advice dynamically as weather conditions change.
4. The system shall allow users to input their skin type (oily, dry, sensitive, etc.) to tailor skincare recommendations.
5. The system shall notify users about extreme weather conditions that could affect their skin (e.g., high UV levels or low humidity).

2.2.6 Module 6: User Interface and Experience Management

Following are the requirements for module 6:

1. The system shall provide a user-friendly interface that is accessible on both mobile and desktop platforms.
2. The system shall allow users to input text-based symptoms and upload images of their skin in an easy-to-use format.
3. The system shall provide clear, step-by-step feedback to guide users through the diagnostic process.
4. The system shall deliver diagnostic results, confidence scores, and skincare recommendations to users in a clear and concise manner.
5. The system shall send notifications to users when new alerts or diagnostic results are available.
6. The system shall store user preferences and past diagnostic results for easy retrieval and ongoing management of skin conditions.

2.3 Non-Functional Requirements

This section defines the quality attributes of the software that are measurable, verifiable, and quantitative.

2.3.1 Reliability

The DiagnoVision system shall maintain a high level of reliability, that will ensure continuous and uninterrupted service for the users seeking diagnoses and medical advice. The system must have a Mean Time Between Failures (MTBF) of at least 1,500 hours for all critical operations, including symptom analysis and image processing. Failures are defined as the inability to process user inputs, diagnose conditions, or display results within the specified performance parameters. The system will implement real-time error detection mechanisms to identify and log failures, followed by automatic correction or prompt notifications to the technical team for resolution. Additionally, redundancy measures will be implemented to ensure minimal disruption in case of failure, providing seamless service continuity.

2.3.2 Usability

Usability requirements focus on enhancing the user experience by ensuring ease of learning, use, error avoidance, efficient interactions, and accessibility for all users. These requirements will guide the user interface designer in creating an optimal experience for the users of varying technical backgrounds.

USE-1: Users shall be able to input symptoms and to upload images within 3 interactions to receive a diagnosis.

USE-2: The system shall provide clear, step-by-step guidance during the diagnosis process to minimize user errors.

USE-3: Users shall receive immediate feedback for any errors in input, allowing for quick corrections without restarting the process.

USE-4: The application shall be navigable within 5 minutes by first-time users, ensuring an intuitive learning curve.

USE-5: The system shall comply with accessibility standards, enabling users with visual impairments to navigate the interface effectively using screen readers.

2.3.3 Performance

Performance requirements for DiagnoVision define the expected operational speed and the efficiency for various system functionalities. Specific performance goals are outlined for critical operations to ensure a responsive user experience.

PER-1: The system shall process an image upload and return a diagnosis within 8 seconds for 95% of the requests.

PER-2: The symptom analysis module shall provide a textual diagnosis based on user input within 6 seconds for 90% of interactions.

PER-4: The chatbot feature shall respond to user queries within 4-5 seconds for 90% of the interactions to facilitate efficient communication.

PER-5: Real-time data retrieval for disease spread notifications shall occur within 10 seconds, providing users timely and relevant information.

2.3.4 Security

The security requirements for DiagnoVision focus on protecting the system and its data from unauthorized access and ensuring user privacy. These requirements are measurable in terms of the effort and skill level needed to breach the system.

To handle security, DiagnoVision will implement the following measures:

SEC-1: The system will be designed to withstand unauthorized access attempts, requiring a significant effort level to compromise user data.

SEC-2: All user data, including personal health information, will be encrypted both at rest and in transit, ensuring that decryption requires advanced skills and substantial time investment.

SEC-3: User accounts will be protected through 2-factor authentication, enhancing security against unauthorized access and ensuring that only verified users can access sensitive data.

SEC-5: Regular security assessments and vulnerability scans will be conducted at least quarterly to identify and address potential security risks, ensuring the ongoing integrity of the system.

2.4 Domain Model

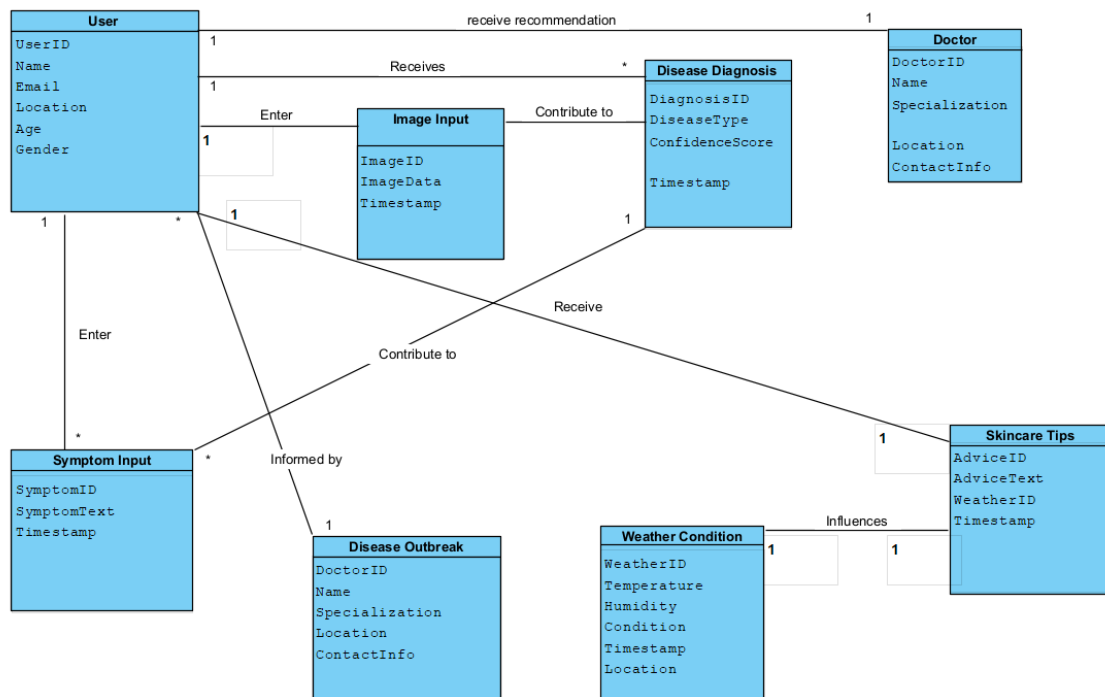


Figure 2.1: Domain Model

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

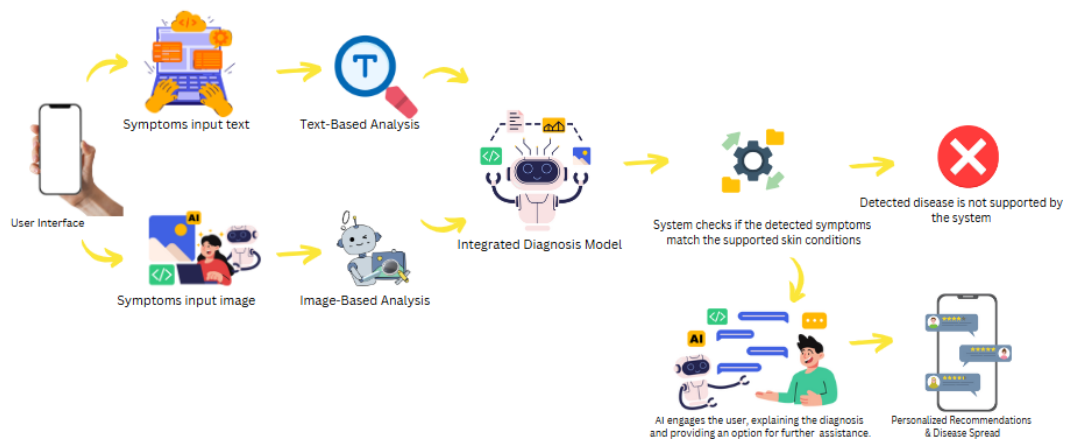


Figure 3.1: Architectural Diagram

The system is designed with a modular architecture to ensure efficient functionality and scalability. Each module has a specific role, from processing user inputs to providing accurate diagnoses and personalized advice. These modules work collaboratively, integrating text and image analysis to deliver reliable health insights. This approach enables the system to function cohesively while remaining adaptable for future enhancements.

3.1.1 Text Input and Analysis Module

Processes user-entered symptoms using natural language processing (NLP) models to identify potential skin conditions based on text-based descriptions.

3.1.2 Image Input and Analysis Module

Analyzes images of skin areas provided by the user, detecting visual patterns indicative of specific skin diseases like Acne, Eczema, and Psoriasis.

3.1.3 Combined Diagnosis Module

Integrates the results from both the text and image modules to generate a more accurate and comprehensive diagnosis using decision-making algorithms.

3.1.4 Disease Spread and Monitoring Module

Tracks disease prevalence in the user's region by analyzing publicly available data, providing insights into local outbreaks and precautions.

3.1.5 Personalized Advice and Dermatologist Recommendation Module

Offers personalized skincare advice based on real-time data and recommends top dermatologists based on the user's condition and location.

3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

Design Models for Procedural Approach

The applicable models for the project using Procedural approach may include:

- Activity Diagram
- Class Diagram
- Class-level Sequence Diagram

3.2.1 Activity Diagram

3.2.2 Class Diagram

3.2.3 System Sequence Diagram

3.3 Data Design

For the *DiagnoVision* project, an Entity Relationship Diagram (ERD) models the key entities and their relationships. Here's a breakdown of the entities based on the report:

Entities: 1.

3.3.1

User - Attributes: UserID, Name, Email, Location, Age, Gender - Relationships: Can input symptoms, upload images, and receive diagnosis.

2.

3.3.2 Symptom Input

- Attributes: SymptomID, SymptomText, Timestamp - Relationships: Belongs to a User, linked to Disease Diagnosis.

3.

3.3.3 Image Input

- Attributes: ImageID, ImageData, Timestamp - Relationships: Belongs to a User, linked to Disease Diagnosis.

4.

3.3.4 Disease Diagnosis

- Attributes: DiagnosisID, DiseaseType (Acne, Eczema, Psoriasis), ConfidenceScore - Relationships: Linked to Symptom Input and Image Input, provides feedback to User.

5.

3.3.5 Doctor

- Attributes: DoctorID, Name, Specialization, Location, ContactInfo - Relationships: Can be recommended to a User based on Diagnosis and Location.

6.

3.3.6 Disease Outbreak

- Attributes: OutbreakID, DiseaseType, Location, Severity, Timestamp - Relationships: Affects the User's recommendations based on Location and Disease.

7.

3.3.7 Weather Condition

- Attributes: WeatherID, Temperature, Humidity, Condition (Rainy, Sunny, etc.), Timestamp - Relationships: Provides skincare advice to User.

8.

3.3.8 Skincare Advice

- Attributes: AdviceID, AdviceText, WeatherID - Relationships: Linked to Weather Condition, tailored to User's diagnosis and current weather.

—

Proposed ERD Structure: - *User* 1—< *Symptom Input* >—1 *Disease Diagnosis* - *User* 1—< *Image Input* >—1 *Disease Diagnosis* - *Disease Diagnosis* >—1 *Doctor* (recommendation based on diagnosis and location) - *User* 1—< *Weather Condition* >—1 *Skincare Advice* - *User* >—1 *Disease Outbreak* (affected by outbreak data in the user's region)

—

3.3.9 Relationships

- **One-to-Many (1—<)**: A User can have multiple symptom inputs, image uploads, diagnoses, etc. - **Many-to-One (>—1)**: Multiple Symptom or Image Inputs result in a single Disease Diagnosis. - **One-to-One**: Diagnosis corresponds to a single doctor recommendation and outbreak/ weather-based advice.

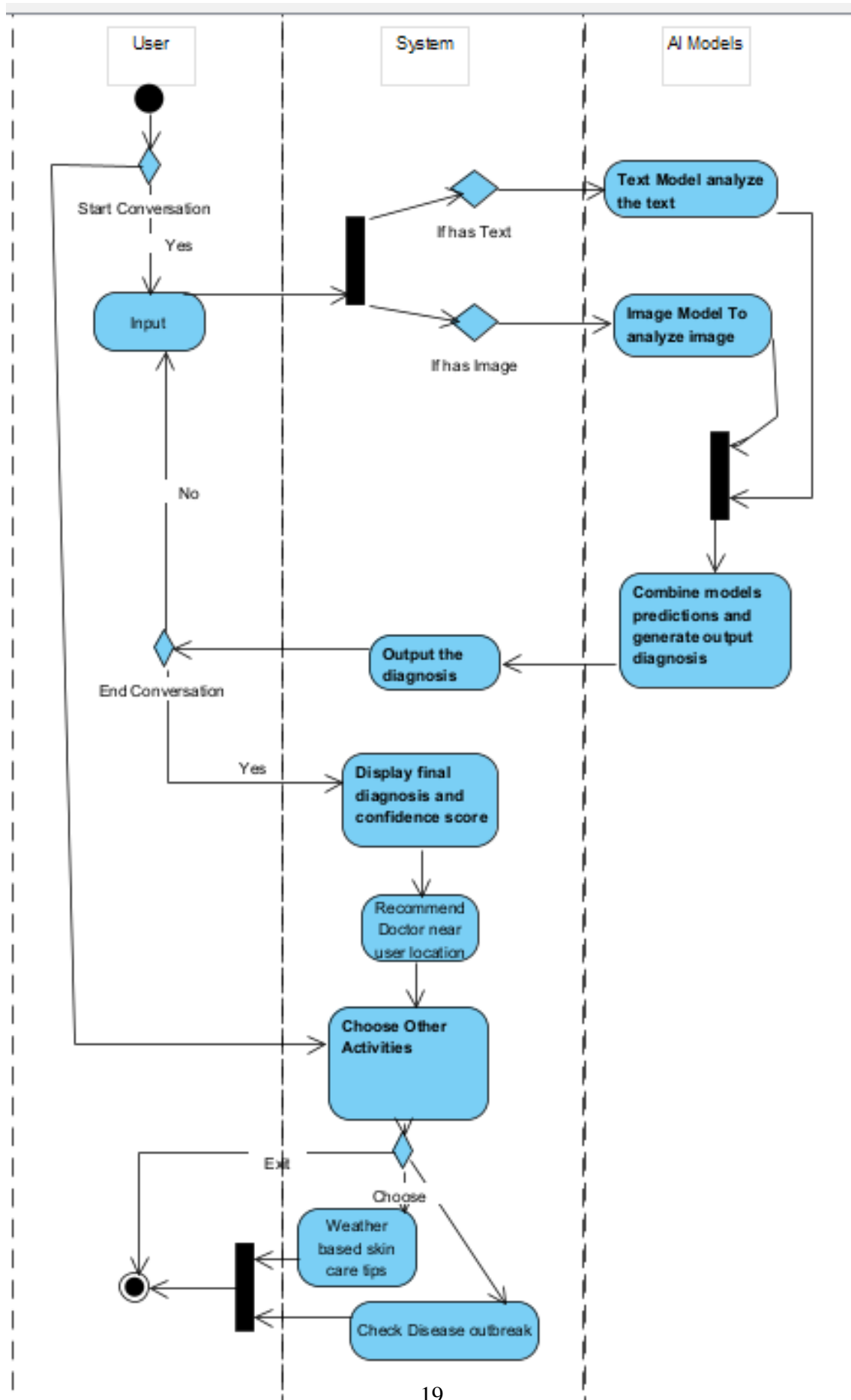


Figure 3.2: Activity Diagram

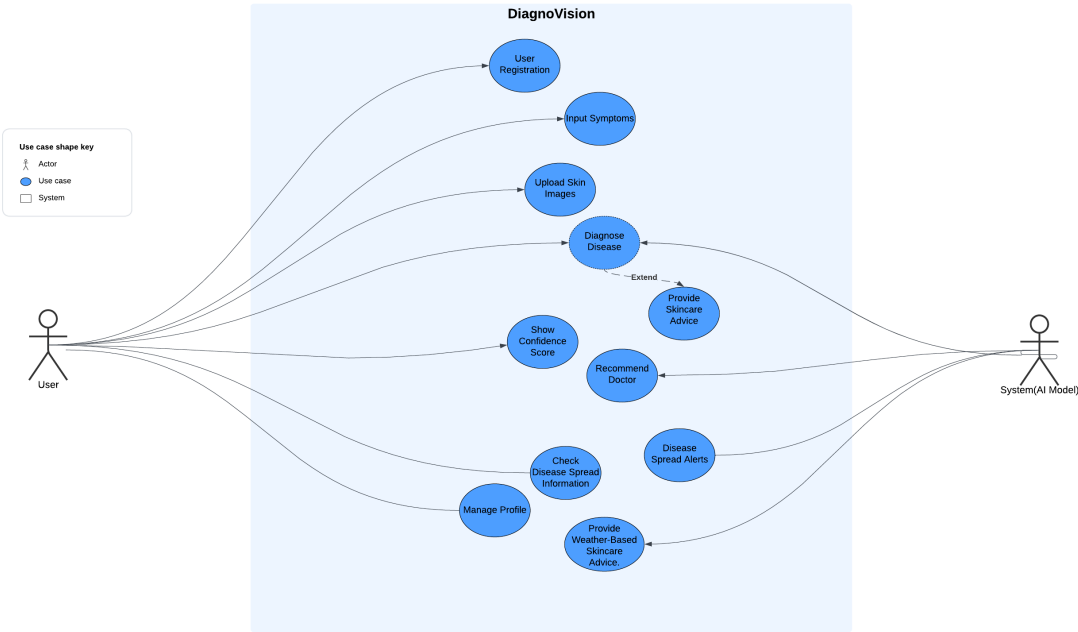


Figure 3.3: Activity Diagram

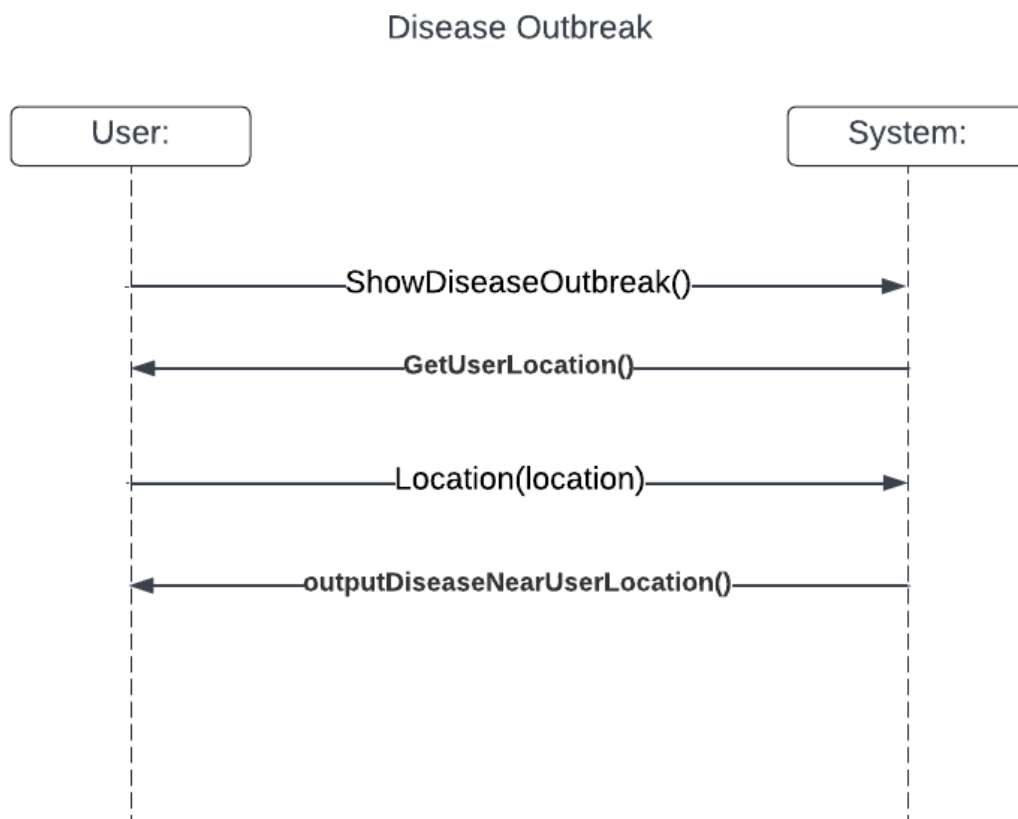


Figure 3.4: SSD

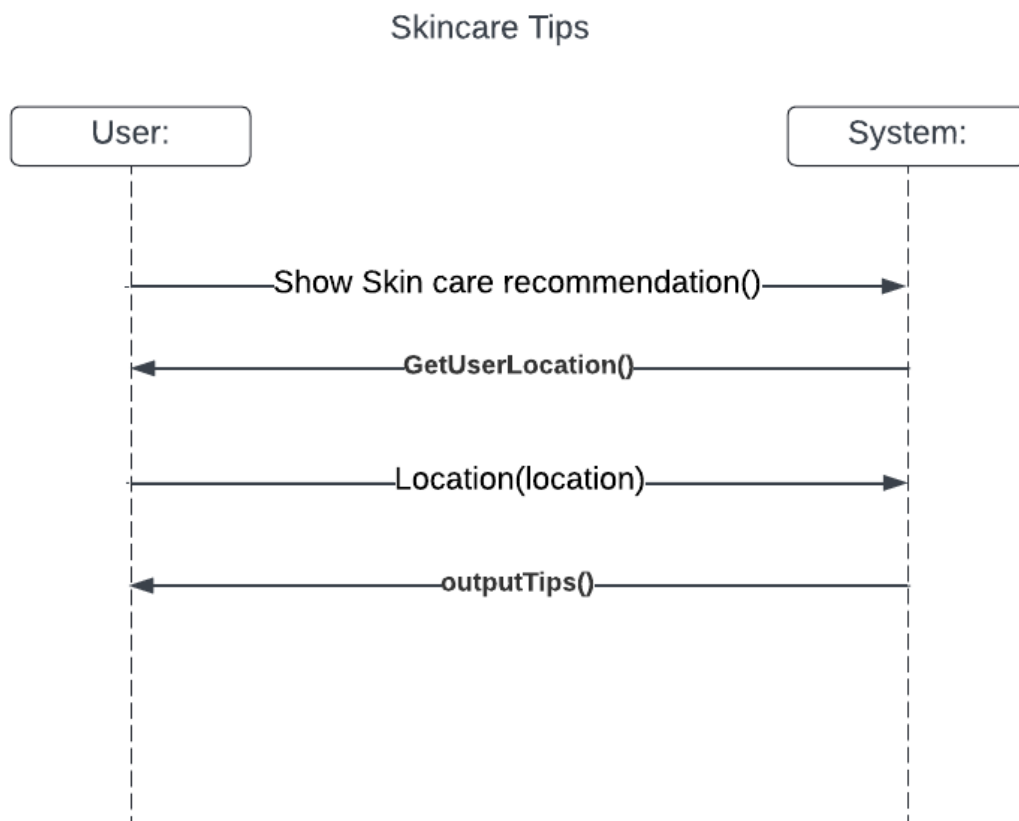


Figure 3.5: SSD

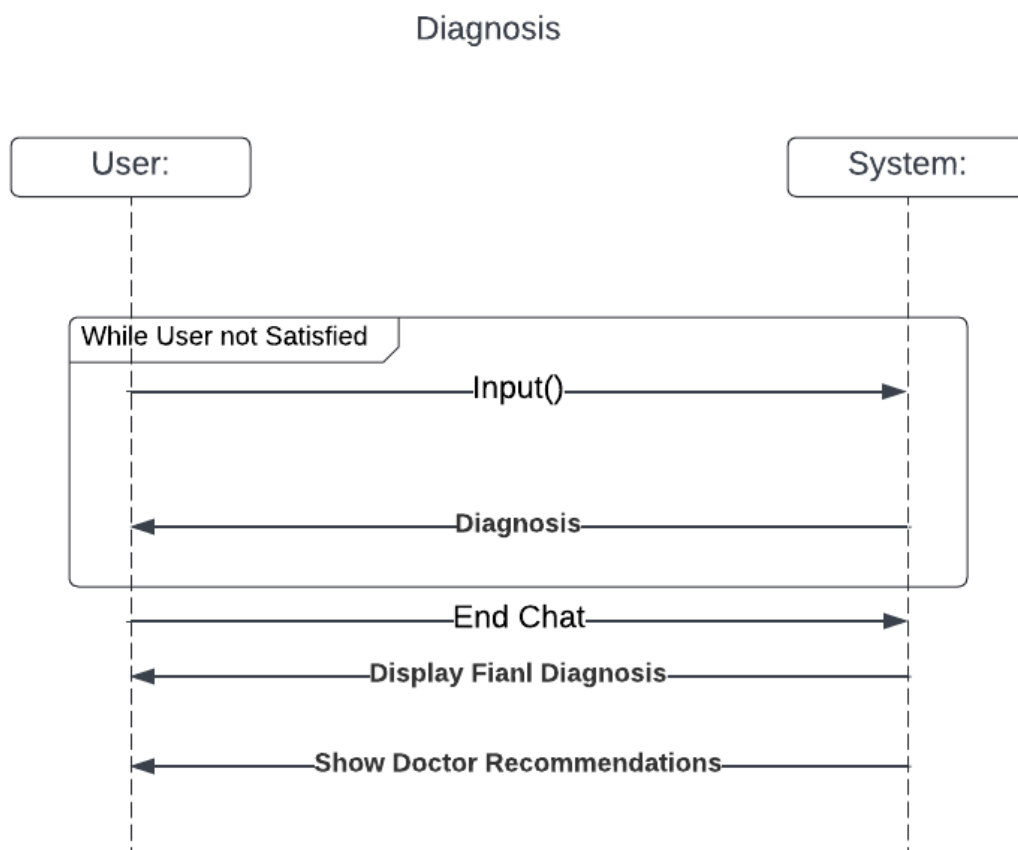


Figure 3.6: SSD

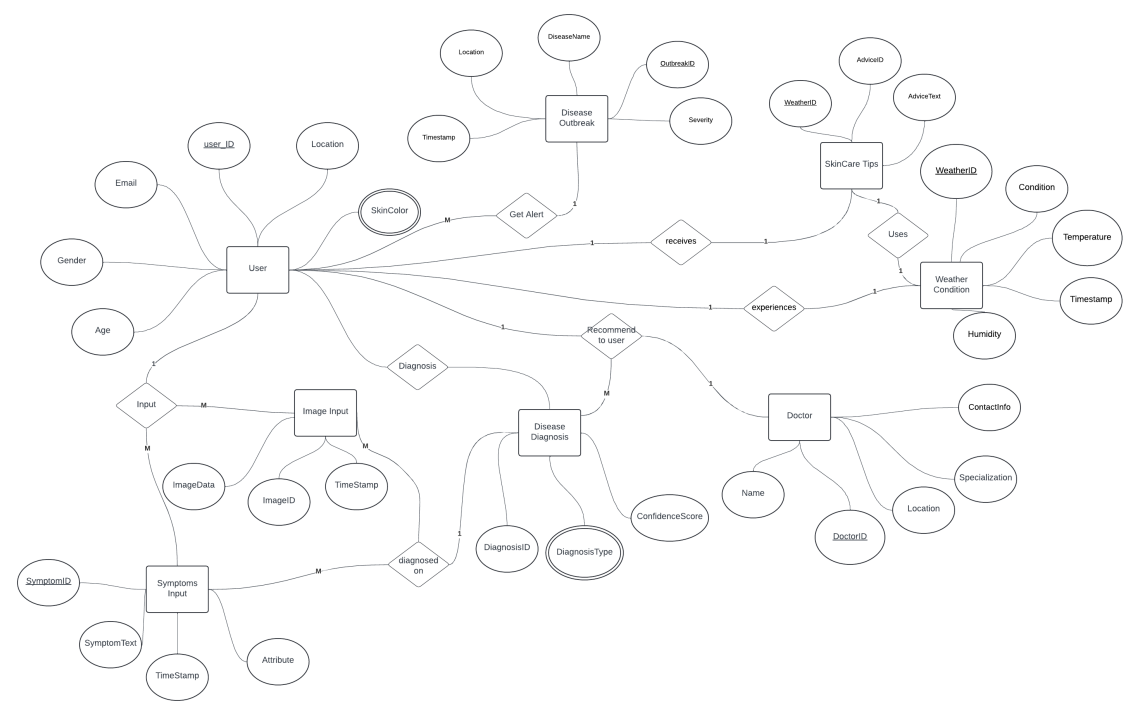


Figure 3.7: Entity Relationship Diagram

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- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.